Methane Oxidation in Swedish Landfills Quantified with the Stable Carbon Isotope Technique in Combination with an Optical Method for Emitted Methane

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Methane budgets (production = emissions + oxidation + recovery) were estimated for six landfill sites in Sweden. Methane oxidation was measured in downwind plumes with a stable isotope technique (Chanton, J. P., et al., *Environ. Sci. Technol.* **1999**, *33*, 3755–3760.) Positions in plumes for isotope sampling as well as methane emissions were determined with an optical instrument (Fourier Transform InfraRed) in combination with N_2O as tracer gas (Galle, B., et al., *Environ. Sci. Technol.* **2001**, *35*, 21–25.) Two landfills had been closed for years prior to the measurements, while four were active. Measurements at comparable soil temperatures showed that the two closed landfills had a significantly higher fraction of oxidized methane (38–42% of emission) relative to the four active landfills (4.6–15% of emission). These results highlight the importance of installing and maintaining effective landfill covers and also indicate that substantial amounts of methane escape from active landfills. Based on these results we recommend that the IPCC default values for methane oxidation in managed landfills could be set to 10% for active sites and 20% for closed sites. Gas recovery was found to be highly variable at the different sites, with values from 14% up to 65% of total methane production. The variance can be attributed to different waste management practices.

Introduction

Methane, one of the most important greenhouse gases, is produced in landfills in large quantities. Waste decomposition has been estimated to account for as much as 17% of the total anthropogenic methane emissions to the atmosphere, or 61 Tg per year (*3*), but there are also arguments for more moderate estimates: around 16–20 Tg (*4*). These estimates are calculated from data on amounts of waste and biogas formation potentials in the different organic fractions of the waste materials. The same approach is used in the IPCC (Intergovernmental Panel for Climate Change) methodology used for estimates of national budgets (*5*).

IPCC's methodology allows the use of default values when reliable experimental data do not exist. This includes the portion of methane that is oxidized by microorganisms in the surface of landfill sites, before it reaches the atmosphere. The default value for methane oxidation has been set to 0% (*5*, *6*), but national reports have also been approved with methane oxidation estimated at 10% (*7*, *8*). However, data from field experiments have hitherto been limited.

For quantitative estimation of the methane oxidation, the isotope method is to our knowledge the only available method for measurements in situ. The method, as first described for landfills (*9*), draws upon the fact that the methane-oxidizing bacteria (methanotrophs) preferentially consume methane containing the common, light isotope ^{12}C and discriminate against methane containing the heavier ^{13}C. Thus, methane oxidation can be estimated by the degree of change between the ^{13}C content of methane emitted from the landfill surface relative to the $\delta^{13}C$ of methane found within the anaerobic part of the landfill. To apply this technique, knowledge of the degree of microbial discrimination, i.e., the fractionation factor α, is required. This factor is measured in laboratory experiments under conditions equivalent to in situ, since the α factor has been found to vary with soil type, temperature, and most likely also the structure of the microbial population.

A further development of the isotope method for the measurement of total landfill methane oxidation is measurement in downwind plumes (*1*), which can be expected to give more representative data for whole landfills than samples of landfill gas accumulated in chambers. Such data will thus be more appropriate for extrapolation of results to models and budgets on national or regional levels.

In a previous study (*2*), we found that the tracer gas technique in combination with a FTIR (Fourier Transform Infrared) instrument for measuring methane emissions from landfills gave highly reproducible data. In this study we also wanted to evaluate this instrument for tracking the landfill gas plume in order to obtain samples for carbon isotope analysis. The major objective for this study was to investigate not only the importance of methane oxidation in the overall methane balance in landfills but also the importance of closing sites with proper covering and to evaluate the effects on methane oxidation caused by an interruption of a gas extraction system. Since many different soils are used for landfill covering, there is always the possibility for introducing inhibitory compounds. Negative effects on methanotrophs due to heavy metals in the cover soils have been reported for rice soils (*10*), and therefore such analyses were made on some of the soils in this study.

Materials and Methods

Landfill Sites. Six landfill sites in different parts of Sweden were investigated in this study (Table 1, Supporting Information). They not only represent different latitudes but also differ in sizes: Filborna is the largest site in Sweden, with shredded household waste deposited in cells located in limited areas (ca. 60 000 tons of household waste per year). Filborna and Hagby have household waste in cells, i.e., limited areas, with horizontal perforated pipelines for gas extraction applied at 10 m or less distance, while the other are "traditional" Swedish landfills, with different types of organic waste deposited on many areas on the site. Two of the landfills

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10.1021/es062735v CCC: $37.00

Published on Web 08/28/2007

were closed before the experiments started: Hagby in 1995 and Visby in 1997, with no more waste materials added after this time. However, landfill gas is extracted and utilized on all sites, leachates are taken care of, etc. The chosen sites also represent different types of cover materials, including mineral soils (Hagby, Visby), mixtures of sewage sludge and soil (Högbytorp, Sundsvall), and mixtures of wood chips and sludge (Filborna, Heljestorp). It should be noted that in the 1970s most of the household waste was burned in Sweden, and all the six landfills were active or started during this period. Despite the lack of reliable statistics before 1994, it seems as if the waste streams have been quite steady, except for the Hagby landfill which received most of its waste in the 1980s and the cumulative amount of waste is probably around 200 ktons.

Methane Emissions. The Time Correlation Tracer system, as described by Galle et al. (*2*), with slight modifications described by Samuelsson et al. (*11*) was used for measuring methane emissions. This includes the release of the tracer gas N_2O from cylinders with controlled flow rates and concentration measurements with an FTIR system. The FTIR instument was built from a Bomem MB104 spectrometer with a resolution of maximum 1 cm^{-1}. The spectrometer has an internal IR-light source equipped with a red laser (HeNe), and it is connected to a 15 L gas cell (Infrared Analysis Inc.) with an optical path adjustable between 3.2 and 107 m, which makes it possible to optimize absorption level according to changes in conditions. The transmitted IR-light is detected with an InSb detector, housed in a temperature controlled box inside a Volkswagen van. An inert pump sucks air from the plume through the gas cell with one volume changed in the cell per recorded spectrum. The measuring system is automatically regulated by a computer, with evaluation and presentation of data in real time. Detection limits for CH_4 was 11.4 ppb and for N_2O 1.0 ppb.

N_2O was released from 2 to 5 cylinders, 15 kg net weight in each, distributed over the landfill surface to match the methane realease distribution, as estimated from an initial on site leak search with the same instrumentation. The amount of released N_2O was determined both by a flow integrator and by weighing the cylinders on a precision scale (±2 g). Depending on the size of the actual landfill site, the amounts of emitted methane and the pattern of the plume, the number of N_2O cylinders on each landfill varied between two and five, and consequently the amount of released N_2O has varied between 5.0 and 12.5 kg h^{-1}. In order to observe the emissions as coming from a point source, sites with large areas demanded that the measurements were conducted at relatively long distances from the site, and on some occasions distances as far as 3 km were used. This implicated an increased tracer gas release to get reliable signals in the plume.

Emissions of methane, E_{CH_4} (kg CH_4 h^{-1}), are determined as

$$E_{CH_4} = F_{N_2O} \cdot [CH_4]/[N_2O] \cdot M(CH_4)/M(N_2O) \qquad (1)$$

where F_{N_2O} is the known release of tracer gas (kg N_2O h^{-1}), [ ] are concentrations in the plume (ppb above background), and M are molecule weights (g mol^{-1}). Background was determined in upwind during stable conditions at 2−3 km distance from the landfills. The ratio of CH_4 to N_2O in eq 1 was obtained from an optimal linear fit of CH_4 versus N_2O for all plume samples. The FTIR instrument was continuously sampling, averaging the plume concentrations for 30−60 s, meaning typically more than 100 samples during the experiment period of 1−4 h. Within that time both the center, the edges of the landfill plume, and the background were sampled. If the ratio was significantly different in the different plume parts, as observed online during the experiment, the tracers were rearranged to better match the methane release.

Cover Soils: Common Characteristics. From the surface of every landfill site at least four individual soil samples of about 1 kg were taken at a depth of 0−30 cm. The samples were treated separately through the analyses (sieve 4 mm, analyses of water content, organic matter, etc.). All of these treatments, including incubations, were done at Linköping University according to methods described earlier (*12*).

Carbon Isotope Analysis. Carbon isotope values are expressed as

$$\delta^{13}C‰ = ((R_{sample}/R_{standard}) - 1) \times 1000 \qquad (2)$$

where R_{sample} is the $^{13}C/^{12}C$ ratio of the sample, and $R_{standard}$ is the $^{13}C/^{12}C$ ratio of a standard marine carbonate (PDB, 0‰).

Fractionation Factors. For determination of α values, incubation temperatures were chosen to correspond with actual field conditions in the investigated soils, i.e., 3, 10, 15, and 20 °C. Thus we determined α-values as a function of temperature for the cover soil of each landfill site.

The soil samples were sieved (4 mm mesh) and stored cold (+3 °C) for a maximum of 5 days before incubation. At the beginning of the experiment, four aliquots (50−100 g ww) of each soil sample were transferred to 1.1-L glass flasks (Schott, Mainz, Germany). The flasks were sealed with gastight screw caps and were then allowed to stand for 1 h at one of four incubation temperatures; 3 °C, 10 °C, 15 °C, and 20 °C (for Filborna 5 °C was used instead of 3 °C). Next, 50 mL of ambient air was inserted, and, shortly thereafter (at time zero), 60 mL of CH_4 was added. Thus the partial pressure of CH_4 was initially 5.0%, which is well over the 1000 ppmv believed to distinguish between high and low affinity methanotrophs (*13*). Zero time samples were taken immediately after the addition of CH_4 after which the flasks were returned to their respective temperatures.

During incubation, triplicate gas samples of 0.3 mL were withdrawn and immediately analyzed on a gas chromatograph with a flame-ionization detector (Packard 428 (*14*)), followed by 10 mL samples removed and stored in pre-evacuated glass vials for later isotopic and concentration analysis. These parallel samples were taken with intervals from 1 h up to 60 days. For every flask, four samples representing a range of methane concentrations over time were chosen for isotopic analysis to determine α. The analyses of $\delta^{13}C$ in incubation samples were made at the Department of Forest Ecology at the Swedish University of Agricultural Sciences in Umeå, according to methods described by Ohlsson and Wallmark (*15*). The fractionation factor, α, is here defined as the ratio of the rate constants for methane containing the light (k_L) and the heavy isotope (k_H) assuming first-order kinetics during depletion (*16*).

$$\alpha = k_L/k_H \qquad (3)$$

In this expression, the lighter isotope reacts at a slightly more rapid rate relative to its concentration than does the heavier isotope because $k_L > k_H$ (*16*). From the ^{13}C-content of methane, the fractionation factors were calculated for each incubation according to methods described by Börjesson et al. (*17*).

Carbon Isotopes in Environmental Samples. Analyses of samples taken in the anoxic zone (e.g., from the gas extraction systems, in gas plumes, and in background upwind air) were made at Florida State University, U.S.A. Some anoxic zone analyses were also made in spring 2001 at University of California, Davis, U.S.A.

Plumes of methane downwind of the landfills were located by the FTIR-instrument, and triplicate samples were collected in 118 mL flasks and sealed with a rubber stopper and aluminum crimp-seal. Fifty milliliters of extra air was immediately added. Corresponding background samples were taken on the upwind side of the landfill. ^{13}C-content of methane in plume samples and upwind samples was analyzed according to methods described by Chanton et al. (*1*). (The number of samples analyzed are given in Table 32). Thus, the excess of $\delta^{13}C$ in the plume was calculated as

$$([\delta CH_4]_{excess} = (([CH_4]_{meas}\, [\delta CH_4]_{meas}) - ([CH_4]_{amb})\, [\delta CH_4]_{amb}))/([CH_4]_{meas} - [CH_4]_{amb}) \quad (4)$$

where $[CH_4]_{meas}$ and $[\delta CH_4]_{meas}$ represent the concentration and $\delta^{13}C$ values of CH_4 in the downwind plume, and $[CH_4]_{amb}$ and $[\delta CH_4]_{amb}$ represent the concentrations and C values of the upwind samples.

The fraction of CH_4 (f_{ox}) oxidized in upward transit through the landfill cover soil is then given by (*18*, *19*)

$$f_{ox} = \frac{(\delta_{excess} - \delta_A)}{1000 \times (\alpha_{ox} - \alpha_{trans})} \quad (5)$$

where δ_{excess} is calculated using eq 4 and δ_A is the carbon isotopic content of anoxic CH_4 from landfill probe data, and α_{ox} and α_{trans} are the isotope fractionation factors appropriate for the soil type (clay or mulch) and associated with transport of CH_4, respectively.

The parameter α_{trans} is assumed to be 1 because transport of CH_4 up through the soil cap is assumed to be dominated by advection, a process that does not cause isotopic fractionation (*9*, *20*). This assumption is supported by observations of a negative relationship between CH_4 emission and atmospheric pressure in several landfills (*21*, *22*), suggesting that, in general, a pressure gradient drives gas flow upward through soil caps. Recent laboratory experiments have shown that this approach can underestimate CH_4 oxidation by not taking into account fractionation associated with methane diffusion (*19*, *23*). Thus isotopic determinations of methane oxidation in field studies are conservative lower limits of methane oxidation. The rate of CH_4 oxidation, M (kg CH_4 h^{-1}), was calculated from flux and percent oxidation using the following equation

$$M = f_{ox}\left(\frac{F}{1 - f_{ox}}\right) \quad (6)$$

where f_{ox} is the fraction oxidized (% oxidized/100), and F is the methane emission rate (kg CH_4 h^{-1}).

Meteorological System. The FTIR and tracer gas system was accompanied by a meteorological system, consisting of two loggers. The main logger was placed in the vicinity of the FTIR system and was used for logging temperature and pressure in the gas cell. The logger was also connected to a 10 m high mast attached to the roof of the bus. Adjacent to the gas inlet at 10 m height on the mast, an anemometer for measuring wind speed and wind direction was mounted, together with sensors for air temperature, barometric pressure, and humidity. This also made it possible for concentration time series to be correlated to the wind direction and thus to different parts of the landfill site. The meteorological data collection was regulated by the FTIR system, and values were stored for every recorded spectrum. The second logger was placed on the top of each landfill site, where data were collected on soil temperature (5 cm depth), air temperature, wind speed, and wind direction at 5 m height. This system was utilized to measure the conditions on the landfill site, enabling the choice of sampling position, i.e., to judge on the direction of the landfill gas plume. It also served as backup for the main logger.

Gas Recovery. Data on gas recovered from the extraction systems were provided by the landfill owners, with analyses based on different principles, with varying time resolution.

TABLE 1. Carbon Isotope Ratios in Landfill Gas Samples

		$\delta^{13}C$		
site	date	no. of samples	$\delta^{13}C$ in CH_4 (‰)	$\delta^{13}C$ in CO_2 (‰)
Filborna	2001-11-15[b]	3	−56.49 (SD 0.27)	nm[c]
Filborna	2002-07-02[b]	3	−57.58 (SD 0.27)	11.47 (SD 0.75)
Heljestorp	2002-05-23[b]	3	−59.77 (SD 0.54)	18.72 (SD 0.62)
Högbytorp	2001-06-07[a]	2	−57.34 (SD 0.11)	nm[c]
Högbytorp	2002-04-11[b]	3	−57.35 (SD 0.10)	7.97 (SD 0.36)
Högbytorp	2003-11-10[b]	5	−59.79 (SD 0.07)	9.77 (SD 0.21)
Sundsvall	2001-06-28[a]	4	−56.02 (SD 0.56)	nm[c]
Sundsvall	2002-03-06[b]	3	−60.17 (SD 0.10)	10.73 (SD 0.30)
Hagby	2001-04-19[a]	2	−59.73 (SD 0.01)	nm[c]
Hagby	2002-04-24[b]	3	−60.37 (SD 0.36)	8.73 (SD 0.77)
Hagby	2003-11-13[b]	5	−61.11 (SD 0.02)	7.35 (SD 0.13)
Visby	2001-06-13[a]	3	−58.30 (SD 0.14)	nm[c]
Visby	2002-06-07[b]	3	−58.51 (SD 0.08)	5.80 (SD 0.55)
Visby	2003-11-26[b]	5	−58.72 (SD 0.10)	4.63 (SD 0.19)

[a] Data from UC Davis. [b] Data from FSU. [c] nm, not measured.

At Filborna, the methane concentration and the gas flow were measured continuously, with an uncertainty of approximately 5%. For Sundsvall, the gas extraction was given as a mean day value based on gas flow and the partial pressure of methane, but the variation over 4 days was less than 2.4%, indicating at least some stability. The same approach was used for Heljestorp and Hagby, supplemented by a couple of manual readings of gas flow and methane ratios in the beginning and in the end of the measurement day. For Högbytorp and Visby the gas extraction was calculated from readings of utilized energy and conversion to methane amounts through the conversion factor 1 kg CH_4 = 16.56 kWh (*24*). Readings were made 1−5 times per day.

Determination of Methane Production. The total methane production P (kg h^{-1}) in the landfills was calculated as

$$P = E + R + M \quad (7)$$

where E = CH_4 emission to atmosphere (kg h^{-1}), R = CH_4 through gas extraction (kg h^{-1}), and M = CH_4 oxidized in the cover (kg h^{-1}).

Results

Cover Soils and Methane Fractionation. The fractionation factor α showed large variations in and between each set of soil replicates (Figure 1, Supporting Information). Although not significant in analysis of variance, the plotted lines for all six soils showed α-values to be higher at low temperatures, as reported by others (*1*, *9*, *17*, *18*). Neither soil moisture (Table 2, Supporting Information) nor methane consumption rates (not shown) seemed to have any influence on α, i.e. no correlation could be observed between these variables and α. The α-values obtained had intercepts (0 °C) between 1.0204 and 1.0358, with plotted lines for the temperature in an interval between −0.000098 K^{-1} and −0.000664 K^{-1} (Figure 1, Supporting Information).

Methane Oxidation in Situ. The carbon isotope ratio ($\delta^{13}C$) of methane in the anoxic zone samples varied between −56.0 and −61.1‰ (Table 1), which reveals that $\delta^{13}C$ was rather constant for each landfill site over time. One exception is Sundsvall, where the greater variability may be due to the fact that the gas extraction system was not operating during the first sampling occasion and so the sample that was obtained may not have been representative.

Background upwind air values generally had more ^{13}C enriched isotope ratios in November (>−46.5‰) than during other periods (<−46.5‰; Table 2).

Calculations of the fraction of methane oxidized showed that the two closed landfills had significantly higher percent-

TABLE 2. Carbon Isotope Ratios in Air Samples from the Landfills and Methane Oxidation Ratios in % of the Sum of Emitted and Oxidized Methane (±SD)

site	date	concn of CH_4 in plume (ppmv)[a]	$\delta^{13}CH_4$ in plume[d]	concn of CH_4 in upwind air (ppmv)[a]	$\delta^{13}CH_4$ in upwind air[d]	excess $\delta^{13}CH_4$ from plume[b]	f_{ox} = fraction of CH_4 oxidized (%)
				Active Sites			
Filborna	2001-11-15	2.56 ± 0.17	−46.86 ± 0.18 (3)	2.01 ± 0.01	−45.30 ± 0.27 (3)	−52.94 ± 1.50	18.4 ± 7.9
Filborna	2001-11-23	3.24 ± 0.24	−48.33 ± 0.21 (2)	nm[e]	nm[e]	−53.44 ± 1.56[c]	15.0 ± 8.0
Filborna	2001-11-28	3.67 ± 1.13	−49.66 ± 1.85 (3)	1.97 ± 0.07	−45.32 ± 0.26 (3)	−55.25 ± 0.52	7.1 ± 2.3
Filborna	2002-07-02	2.44 ± 0.06	−48.55 ± 0.04 (3)	1.85 ± 0.02	−47.00 ± 0.08 (3)	−53.39 ± 0.29	22.1 ± 1.5
Heljestorp	2002-05-23	3.48 ± 0.54	−49.58 ± 0.57 (3)	2.11 ± 0.01	−47.02 ± 0.08 (3)	−53.65 ± 0.14	24.8 ± 0.6
Högbytorp	2002-04-11	3.08 ± 0.17	−49.63 ± 0.63 (3)	2.07 ± 0.07	−46.43 ± 0.23 (3)	−56.18 ± 0.96	6.0 ± 4.9
Högbytorp	2003-11-10	4.80 ± 0.52	−52.11 ± 0.75 (3)	2.28 ± 0.09	−45.55 ± 0.39 (5)	−58.09 ± 0.64	7.7 ± 2.9
Sundsvall	2002-03-04	4.90 ± 1.23	−52.30 ± 1.36 (3)	2.10 ± 0.01	−46.65 ± 0.06 (2)	−56.83 ± 0.43	14.5 ± 1.9
				Closed Sites			
Hagby	2002-04-24	8.12 ± 2.23	−50.86 ± 0.57 (3)	1.99 ± 0.04	−47.13 ± 0.10 (3)	−52.18 ± 1.24	36.7 ± 5.6
Hagby	2003-11-13	3.21 ± 0.67	−47.46 ± 0.94 (4)	2.31 ± 0.13	−46.29 ± 1.74 (4)	−50.25 ± 1.05	42.8 ± 4.1
Visby	2003-11-26	4.79 ± 0.74	−48.50 ± 0.65 (5)	2.27 ± 0.09	−45.44 ± 0.24 (5)	−51.39 ± 1.38	37.9 ± 7.1

[a] The same number of samples as $\delta^{13}CH_4$ in the next column. [b] The same number of samples as $\delta^{13}CH_4$ in the plumes. [c] Background from Nov 15, 2001 used. [d] The number of samples are in parentheses. [e] nm, not measured.

TABLE 3. Data from the Landfill Measurements 2001−2003

site	date	soil temp at 5 cm depth (°C)	E = CH_4 emission to atmosphere (kg CH_4 h^{-1})	R^2 for E	R = gas recovery (kg CH_4 h^{-1})	M = CH_4 oxidation (kg CH_4 h^{-1})	P = total CH_4 production (E+R+M) (kg CH_4 h^{-1})	ratio of CH_4 emitted to atmosphere (E/P) (%)	ratio of gas recovery (R/P) (%)
					Active Sites				
Filborna	2001-11-15	9.5	385	0.94	832	86.8	1304	30	64
Filborna	2001-11-23	3.0	441	0.82	820	77.8	1339	33	61
Filborna	2001-11-28	2.7	nm[a]	-	closed	-	-	-	-
Filborna	2002-07-02	13.9	346	0.80	806	98.1	1250	28	64
Heljestorp	2002-05-23	17.0	191	0.82	262	63.0	516	37	51
Högbytorp	2002-04-11	7.3	393	0.96	202	25.1	620	63	33
Högbytorp	2003-11-10	4.9	382	0.84	291	31.9	705	54	41
Sundsvall	2002-03-09	−1.9	33.8	0.50	58.3	5.7	98	35	59
					Closed Sites				
Hagby	2002-04-24	14.2	124	0.98	32.0	71.9	234	53	14
Hagby	2003-11-13	3.0	141	0.97	65.7	105.5	312	45	21
Visby	2003-11-26	5.1	10.7	0.88	32.4	6.5	50	21	65

[a] nm, not measured.

ages of oxidized methane (36.7−42.8%) than the four active landfills (6.0−24.8%; Table 2). This difference is obvious also when comparing the excess δ^{13}C-values (Table 2). It should be noted that most of the measurements were done at comparable soil temperatures in November (Table 3).

The Filborna landfill had the gas extraction system turned off during the measurement November 28, 2001. If we assume that all of the gas normally recovered was released and added to the normal emissions, the amount of methane oxidized can be calculated from the f_{ox}-value of 7.1% in Table 2, arriving at around 90 kg of methane oxidized, i.e., at the same level as when the gas extraction system was working as normal.

There was a positive correlation between soil temperature and methane oxidation in the active landfills (p=0.035 for a one-tailed test) but not for the closed landfills (Figure 1).

Methane Emissions. Methane emissions from the investigated landfills ranged from 10.7 to 441 kg h^{-1}. Generally, r^2-values of 0.80−0.98 were obtained for the measurements (cf. Table 3), with one exception—the measurement at Sundsvall where r^2 = 0.50 owes to a complex topography, with large differences in height between the site and the measuring position. The r^2-value here corresponds to the optimal linear fit (coefficient of determination) to all plume samples of CH_4 versus N_2O during the experiment, typically more than 100 samples.

Table 3 shows results from the measurements on the six landfill sites, together with data on gas recovery from the respective landfill companies.

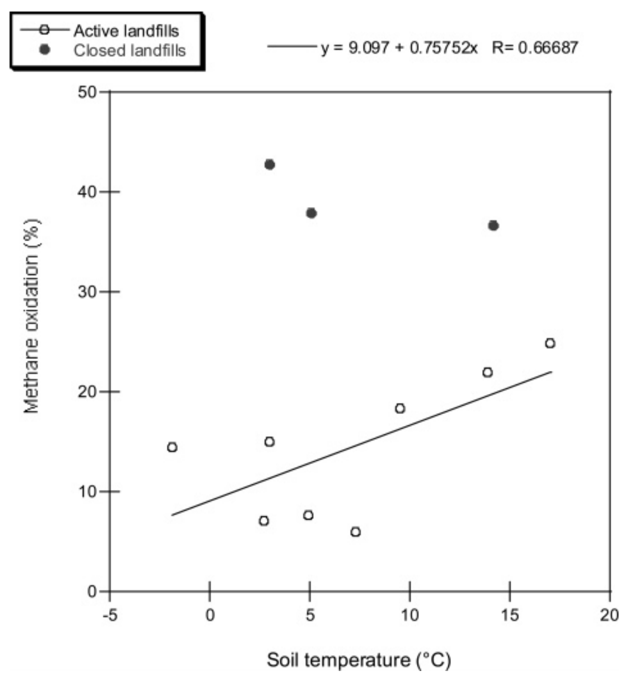

FIGURE 1. Correlation between soil temperature and fraction of methane oxidized. Linear regression is for active landfills only (r^2=0.35, p = 0.035 in a one-tailed test).

At the Hagby site, the gas extraction system was not working properly during any of the two measurements reported here. For comparison, the emission rate measured in 2001 with the same equipment was only 49 kg of CH_4 h^{-1} (*11*). If we assume an oxidation ratio in 2001 similar to the 37% obtained in 2002, the ratio of methane emitted to the atmosphere (*E*/*P*, cf. Table 3) would be 21%, which is more in line with the 24% estimated for the other closed site, Visby.

The ratio of emitted to produced methane varied from 21 to 63%. Landfill size did not affect this ratio (cf. Table 1, Supporting Information). More remarkable was that Högbytorp emits methane at almost the same level as Filborna, although the volume of recovered gas was almost four times lower (Table 3). According to the operator of Högbytorp, Ragnsells (Hellström, N., personal communication, 2003-01-14), several steps were taken to reduce methane emissions during late 2002, such as closure of some areas, additional covering of other areas, replacement of old fan blowers, etc. A minor positive effect on the efficiency of the gas recovery could possibly be detected as the ratio of emitted to produced methane decreased from 63 to 54% in November 2003 (Table 3).

Total Methane Production. The total methane production varied from 50 to 1339 kg CH_4 h^{-1} and was highest at Filborna (Table 3). The variation within the landfills sites was quite low, indicating both that the measurements were reliable and that the conditions for methane production were constant.

Discussion

Methane Emissions. The FTIR methodology proved to work well regardless of climatic conditions and the size of the landfills (Table 3 and Table 1, Supporting Information).

Methane Oxidation. The fact that the isotope ratios were higher in background upwind air samples in November (over −46.5‰; Table 2) than in those collected during the other periods follows the seasonal pattern in mixing ratios observed by others, e.g., in Great Britain (*25*), where $\delta^{13}C$ in methane ranged between a maximum of −47.10‰ in December 1996 to a minimum of −47.50‰ in September 1997. This emphasizes the importance of collecting background samples on all occasions.

The variation of $\delta^{13}C$ in anaerobic zone methane at −56.0 to −61.1% was smaller compared to other reports, e.g., ref *1*, where most values for $\delta^{13}C$ in methane were between −50.2 and −61.3‰. Roughly, $\delta^{13}C$ in anaerobic zone methane could be a measure of the degradation status, where a more negative value would show that organic matter degradation has occurred over a longer time as methane production in the landfill evolved from a greater relative importance of acetate fermentation to more CO_2 reduction (*26*, *27*). This hypothesis is contradicted, both by data in Table 3 and by data from the Hökhuvud landfill, which had been closed 17 years before the samples were taken in 1997, where the $\delta^{13}C$ in methane was −58.6 ‰ (*17*), and which does not exhibit the most negative values (cf. Table 1). Anaerobic zone samples were taken from the gas collection systems, and, since most of this gas is drained from the upper parts of the landfills, we believe that these values should be representative for the gas that enters the surface from below the oxidation zone.

The weakest part of the carbon isotope approach for the determination of methane oxidation is the fractionation factor α. Due to the high variation in α-values, the estimates of methane oxidation in situ must be taken with some caution. Most of the α-values determined in this study were lower than what has been reported by other landfill studies (cf. ref *1*, where α-values lie between 1.0240 and 1.0316). This can be due to differences in type of soil and diffusion as well as to the structure of the methane-oxidizing populations and methanotroph cell densities (cf. ref *28*). However, regardless of the α-values and the calculated methane oxidation, it is important to notice that both closed landfill sites were observed to have significantly more positive excess CH_4 $\delta^{13}C$ values than any of the active sites (one way analysis of variance with data in Table 2).

There was no clear difference between soil types in this study, which seems to contradict the observations by Chanton and Liptay (*18*), who used the isotope method on chamber measurements in order to compare different cover materials on a landfill site in Florida and found that methane oxidation in a mulch topsoil was higher than a clay soil. In this context it is also important to point out that the in situ methane oxidation rates obtained with the isotopes in our study were not comparable to the methane oxidation capacity that was shown during the in vitro incubations. This emphasizes the difference between the system in a closed flask and the conditions prevailing in situ, which is an open system where methane can easily escape. The methane oxidation capacities in three of the investigated soils, Filborna, Sundsvall and Visby, have been reported earlier (*12*), where it was found that samples from Sundsvall and Visby had values around four times higher than Filborna (e.g., 0.55 and 0.46 compared to 0.11 μmol CH_4 g dw soil^{-1} h^{-1} at 10 °C). Data for the other three landfills in this study (not reported) were comparable to Filborna. In this study as well as in others (e.g., refs *29* and *30*), incubations show high consumption rates for soil samples rich in organic matter (in this case Sundsvall), while the mineral soils generally show low consumption rates in flasks (e.g., the Hagby samples in this study). Our study shows that this is not always comparable to field conditions, and the ability of mineral soils for methane oxidation may be severely underestimated, and that a tight clay in combination with a good system for gas recovery is the best way to avoid methane emissions at high rates.

We did not include a factor for isotopic fractionation of methane associated with molecular diffusion. In calculating the fraction of methane oxidized with eq 5, we assumed that $\alpha_{trans} = 1$. Discussions concerning the validity and resulting sensitivity of the results of this assumption has recently been published (*19*, *23*). Allowing for diffusive fractionation clearly increases % oxidation. For example, increasing the fractionation for diffusion from 0‰ to 10‰ (α_{trans} to 1.01) changes the % oxidized from 18.4 to 31.5% for Filborna, Table 3 row 1. However, there is no method to apportion methane transport from a landfill between advection and diffusion. We are presently confident that our results represent a lower limit determination of methane oxidation. At minimum, we know the direction of our error. Were we to increase methane oxidation by estimating diffusive transport, we would no longer be as certain as to the sign of our error. De Visscher et al. (*19*) concluded that the assumption of $\alpha_{trans} = 1$ could result in methane oxidation being underestimated by a factor 2−4. With oxidation rates at over 40% a factor of 2−4 is hardly applicable to our results, and the reason for assuming that molecular diffusion fractionation is of minor importance in our study is also that we believe that wind was responsible for much of the advective transport (2−10 ms^{-1} during the experiments), in addition to the pressure inside the landfills.

Despite high variation in methane oxidation values, it seems obvious that the covers on the closed sites (Visby and Hagby) oxidized a greater proportion of methane than the active sites. The difference between these two categories could be due not only both to wounds in the cover of the active landfills, where large quantities of methane are released without much cover detaining, but also possibly to high methane production in fresh garbage in the active landfills.

In the Filborna series, the highest proportion of oxidized methane was observed in July 2002, which can be explained by the oxidation process being enhanced by high temperatures (e.g., refs *14* and *18*). There was a positive correlation between soil temperature and methane oxidation (Figure 1), that it is relatively weak could be attributed to oxidation normally occurring deeper in the cover than 5 cm (*30*).

The interruption of the gas extraction system made at Filborna in November 2001 had no significant effect on the methane oxidation. Probably a longer interruption would have been needed in order to discover a significant increase in the ratio of methane oxidized.

Total Methane Production and Gas Recovery. If we assume steady-state conditions, which means that the waste amounts deposited each year and its biogas production potential is the same over time, we can compare the methane production values in Table 3 with amounts of household waste in Table 1, Supporting Information. This would give 185 kg of CH_4 per ton household waste for Filborna, 187 for Högbytorp, 160 for Heljestorp, 113 for Sundsvall, 120 for Hagby, and 35 kg of CH_4 per ton household waste for Visby. This is similar to the normal midrange of 120 kg of CH_4 per ton municipal solid waste calculated by Themelis and Ulloa (*31*).

The main reason for the difference in gas recovery between the landfills is most likely due to differences in management practices, where the system applied at Filborna with shredded household waste in cells and narrow distance between pipelines seems to be more effective than the others. A gas recovery, similar to Filborna, of 70% of the total methane production may therefore be regarded as an optimum. Higher gas collection was reported for French landfills by Spokas et al. (*32*), but their efficiency rates at over 90% may be overestimates, since the flux measurements with SF_6 tracer measurements were done on the edge of the landfill rather than at some distance (Morcet, M., personal communication, 2003.). The tracer approach flux determination requires both that the tracer and the gas of interest are well mixed, which requires that measurements are made at some distance, and that the tracer covers the source, which most often requires a number of tracer gas release points. In Visby, it was found that a distance of 1030 m was appropriate for the lowest possible standard deviation, which was a distance ca. five times the landfill diameter (For a detailed test protocol see ref *33*.). In a full plume transect approach, the tracer and the source plume do not need to be matched along the transect direction. The plume transport time for the tracer and the methane plume from the source to the sampling location must however be similar. The closer to the source that sampling is done, the less plume transport time, and the more sensitive the result gets to tracer-source misplacement in the direction orthogonal to the sample transect. The chamber determined rates in the Spokas et al. (*32*) study may also be an underestimate since the coarse chamber grid of 20×20 m which was used will inevitably underestimate emissions, by perhaps as much as a factor of 10 (cf. ref *34*).

The increase in total methane production observed for the closed Hagby site between the years (Table 3) points at a need for longer series of measurements, to enable predictions of the duration of methane emissions from closed landfills.

If a default value for methane oxidation in active landfills is set to 10% (cf. ref *35*), a corresponding value for closed landfills, covered with at least 1 m soil according to the laws in most countries, could be 20%, still a value on the conservative side. A more severe problem is how to estimate the efficiency of the different gas recovery systems, which calls for more studies in order to categorize them.

Acknowledgments

This research was financially supported by the Swedish National Energy Administration, contract P-10856. The time that Chanton spent on this work was supported by the U.S.A.'s National Science Foundation (NSF) under Grant No. 0093677. Cooperative efforts from the people at the landfill sites is highly acknowledged. We thank the anonymous reviewers for improving the manuscript.

Supporting Information Available

Landfill sites (Table 1), characteristics for investigated landfill cover soil samples (Table 2), and linear regression analysis of the fractionation factor α as a function of temperature for six landfill cover soils (Figure 1). This material is available free of charge via the Internet at http://pubs.acs.org.

Literature Cited

(1) Chanton, J. P.; Rutkowski, C. M.; Mosher, B. Quantifying methane oxidation from landfills using stable isotope analysis of downwind plumes. *Environ. Sci. Technol.* **1999**, *33*, 3755−3760.

(2) Galle, B.; Samuelsson, J.; Svensson, B. H.; Börjesson, G. Measurements of methane emissions from landfills using a time correlation tracer method based on FTIR absorption spectroscopy. *Environ. Sci. Technol.* **2001**, *35*, 21−25.

(3) Wuebbles, D. J.; Hayhoe, K. Atmospheric methane and global change. *Earth Sci. Rev.* **2002**, *57*, 177−210.

(4) Bogner, J.; Matthews, E. Global methane emissions from landfills: New methodology and annual estimates 1980−1996. *Global Biogeochem. Cycles* **2003**, *17*, 1065, doi:10.1029/2002GB001913.

(5) IPCC. Revised 1996 *IPCC Guidelines for National Greenhouse Gas Inventories. Workbook (Volume 2). Waste.* http://www.ipcc-nggip.iges.or.jp/public/gl/invs5e.htm (accessed June 7, 2006).

(6) CDM (Clean Development Mechanism). *Indicative simplified baseline and monitoring methodologies for selected small-scale CDM project activity categories.* http://cdm.unfccc.int/EB/Meetings/023/eb23_repan21.pdf (accessed June 7, 2006).

(7) Frøiland Jensen, J. E. *Beregningsmodell for utslipp av metangass fra norske deponier* (In Norwegian); Report 99:16; Statens Forurensningtilsyn: Oslo, Norway, 1999.

(8) Zuber, A.; Tingstorp, S.; Bergstedt, E.; Jansbo, K.; Lundberg, S.; Adolfsson, R.; Börjesson, G. *Framtida metanemissioner från deponier. Underlag för Sveriges nationalrapport till Klimatkonventionen (Future methane emissions from landfills. Basis for Sweden's national report to the Climate Convention)* (in Swedish); Report 5169; Naturvårdsverket (Swedish Environmental Protection Agency): Stockholm, Sweden, 2001; 28 pp.

(9) Liptay, K.; Chanton, J.; Czepiel, P.; Mosher, B. Use of stable isotopes to determine methane oxidation in landfill cover soils. *J. Geophys. Res., [Atmos.]* **1998**, *103* (D7), 8243−8250.

(10) Mohanty, S. R.; Bharati, K.; Deepa, N. Influence of heavy metals on methane oxidation in tropical rice soils. *Ecotoxicol. Environ. Saf.* **2000**, *47*, 277−284.

(11) Samuelsson, J.; Börjesson, G.; Galle, B.; Svensson, B. H. *The Swedish landfill methane emission project*, Proceedings Sardinia 2001, Eighth International Waste Management and Landfill Symposium, CISA Publishers: Cagliari, Italy, 2001; Vol II., pp 485−494.

(12) Börjesson, G.; Sundh, I.; Svensson, B. H. Microbial oxidation of CH_4 at different temperatures in landfill cover soils. *FEMS Microbiol. Ecol.* **2004**, *48*, 305−312.

(13) Bender, M.; Conrad, R. Effect of CH_4 concentrations and soil conditions on the induction of CH_4 oxidation activity. *Soil Biol. Biochem.* **1995**, *27*, 1517−1527.

(14) Börjesson, G.; Svensson, B. H. Seasonal and diurnal methane emissions from a landfill and their regulation by methane oxidation. *Waste Manage. Res.* **1997**, *15*, 33−54.

(15) Ohlsson, K. E. A.; Wallmark, P. H. Novel calibration with correction for drift and non-linear response for continuous flow isotope ratio mass spectrometry applied to the determination of delta N-15, total nitrogen, delta C-13 and total carbon in biological material. *Analyst* **1999**, *124*, 571−577.

(16) Hoefs, J. *Stable Isotope Geochemistry*; Springer-Verlag: Berlin, Germany, 1997.

(17) Börjesson, G.; Chanton, J.; Svensson, B. H. Methane oxidation in two Swedish landfill covers measured with carbon-13 to carbon-12 isotope ratios. *J. Environ. Qual.* **2001**, *30*, 369−376.

(18) Chanton, J.; Liptay, K. Seasonal variation in methane oxidation in a landfill cover soil as determined by an in situ stable isotope technique. *Global Biogeochem. Cycles* **2000**, *14*, 51−60.

(19) De Visscher, A.; De Pourcq, I.; Chanton, J. Isotope fractionation effects by diffusion and methane oxidation in landfill cover soils. *J. Geophys. Res., [Atmos.]* **2004**, *109* (D18111), doi:10.1029/2004JD004857.

(20) Bergamaschi, P.; Lubina, C.; Königstedt, R.; Fischer, H.; Veltkamp, A. C.; Zwaagstra, O. Stable isotope signatures (δ13, δD) of methane from European landfill sites. *J. Geophys. Res.* **1998**, *103* (D7), 8251−8265.
(21) Czepiel, P. M.; Mosher, B.; Harriss, R. C.; Shorter, J. H.; McManus, J. B.; Kolb, C. E.; Allwine, E.; Lamb, B. K. Landfill methane emissions measured by enclosure and atmospheric tracer methods. *J. Geophys. Res.* **1996**, *101* (D11), 16,711−16,719.
(22) Czepiel, P. M.; Shorter, J. H.; Mosher, B.; Allwine, E.; McManus, J. B.; Harriss, R. C.; Kolb, C. E.; Lamb, B. K. The influence of atmospheric pressure on landfill methane emissions. *Waste Manage.* **2003**, *23*, 593−598.
(23) Powelson, D.; Chanton, J.; Abichou, T. Methane oxidation in biofilters measured by mass-balance and stable isotope methods. *Environ. Sci. Technol.* **2007**, *41*, 620−625.
(24) Gendebien, A.; Pauwels, M.; Constant, M.; Ledrut-Damanet, M.-J.; Nyns, E.-J.; Willumsen, H.-C.; Butson, J.; Fabry, R.; Ferrero, G.-L. *Landfill Gas - From Environment to Energy*; Report EUR 14017/1; Commission of the European Communities: Luxembourg, 1992.
(25) Lowry, D.; Holmes, C. W.; Rata, N. D.; O'Brien, P.; Nisbet, E. G. London methane emissions: Use of diurnal changes in concentration and delta C-13 to identify urban sources and verify inventories. *J. Geophys. Res., [Atmos.]* **2001**, *106* (D7), 7427−7448.
(26) Schoell, M. The hydrogen and carbon isotopic composition of methane from natural gases of various origins. *Geochim. Cosmochim. Acta* **1980**, *44*, 649−661.
(27) Whiticar, M. J. Carbon and hydrogen isotope systematics of bacterial formation and oxidation of methane. *Chem. Geol.* **1999**, *161*, 291−314.
(28) Templeton, A. S.; Chu, K.-H.; Alvarez-Cohen, L.; Conrad, M. E. Variable carbon isotope fractionation expressed by aerobic CH_4-oxidizing bacteria. *Geochim. Cosmochim. Acta* **2006**, *70*, 1739−1752.
(29) Nozhevnikova, A.; Lifshitz, A. B.; Lebedev, V. S.; Zavarzin, G. A. Emission of methane into the atmosphere from landfills in the former USSR. *Chemosphere* **1993,** *26*, 401−417.
(30) Börjesson, G.; Sundh, I.; Tunlid, A.; Svensson, B. H. Methane oxidation in landfill cover soils, as revealed by potential oxidation measurements and phospholipid fatty acid analyses. *Soil Biol. Biochem.* **1998**, *30*, 1423−1433.
(31) Themelis, N. J.; Ulloa, P. A. Methane generation in landfills. *Renewable Energy* **2007**, *32*, 1243−1257.
(32) Spokas, K.; Bogner, J.; Chanton, J. P.; Morcet, M.; Aran, C.; Graff, C.; Moreau-Le Golvan, Y.; Hebe, I. Methane mass balance at three landfill sites: What is the efficiency of capture by gas collection systems? *Waste Manage.* **2006**, *26*, 516−525.
(33) Samuelsson, J.; Galle, B.; Börjesson, G. *Metan från avfallsupplag i Sverige (Methane from landfills in Sweden)* (in Swedish); Final report STEM project P10856-4; Energimyndigheten (The Swedish Energy Agency): Eskilstuna, Sweden, 2005; 53 pp.
(34) Börjesson, G.; Danielsson, Å.; Svensson, B. H. Methane fluxes from a Swedish landfill determined by geostatistical treatment of static chamber measurements. *Environ. Sci. Technol.* **2000**, *34*, 4044−4050.
(35) IPCC 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 5 Waste.* http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol5.htm (accessed Dec 22, 2006).

Received for review November 15, 2006. Revised manuscript received May 18, 2007. Accepted July 11, 2007.

ES062735V